
EVALUATING 2D AND 3D U-NET ARCHITECTURES FOR MULTIMODAL BRAIN TUMOR SEGMENTATION

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Accurate segmentation of brain tumors is vital for clinical treatment planning, yet manual annotation remains a time-consuming challenge. While U-Net architectures are the standard solution for this task, finding the optimal balance between spatial context and computational efficiency remains an open question. This study systematically compares 2D, 2.5D, and 3D U-Net models on the BraTS 2020 dataset to evaluate how different levels of spatial information affect performance. Our results indicate that the 2.5D U-Net achieved the best performance (Dice score ≈ 0.80), successfully leveraging local volumetric context to improve boundary detection. Conversely, the full 3D U-Net yielded the lowest performance (Dice score ≈ 0.56). This unexpected result was attributed to optimization instability caused by the small batch sizes and reduced model capacity required to fit the full 3D volumes into memory. We conclude that efficient slice-aggregation strategies, such as the 2.5D U-Net, currently offer a robust and practical solution.

Keywords Deep Learning · Brain Tumor Segmentation · BraTS Challenge · U-Net

1 Introduction

Brain tumors, particularly gliomas, represent one of the most challenging conditions in oncology due to their heterogeneous appearance, complex morphology, and infiltrative growth patterns [Comba et al., 2021]. Accurate segmentation of tumor sub-regions in magnetic resonance imaging (MRI) is essential for clinical decision-making, including surgical planning, radiotherapy target definition, and longitudinal monitoring of treatment response. However, manual segmentation by expert radiologists is labor-intensive, requiring hours per case, and exhibits considerable inter- and intra-observer variability. These limitations create a significant bottleneck in clinical workflows and large-scale research studies.

Recent advances in deep learning, particularly convolutional neural networks (CNNs), have demonstrated remarkable success in medical image segmentation tasks. CNNs have achieved advanced results in various medical segmentation tasks [He et al., 2015], [Huang et al., 2016], [Ronneberger et al., 2015]. Central to their operation is the use of filter kernels that traverse the input image. These kernels contain learnable weights, which are multiplicatively applied to image pixels to produce a single output value. This allows the network to learn relevant features in the input image [Zhang et al., 2021]. The fully convolutional U-Net architecture has emerged as the de facto standard for biomedical image segmentation due to its ability to capture both local features and global context through an encoder-decoder structure with skip connections [Yin et al., 2022].

Despite the strong performance of U-Net-based architectures, several open research questions remain. A key issue is how different network architectures (2D, 2.5D, and 3D) balance spatial context, computational requirements, and overall segmentation accuracy. While 2D models are computationally efficient, they lack volumetric information; 3D models capture full context but require significantly more resources; and hybrid or multi-view approaches attempt to bridge this gap. The 2.5D approach offers a middle ground by aggregating information from multiple consecutive slices while still using computationally efficient 2D convolutions, thereby incorporating limited 3D context without the memory overhead of complete volumetric processing.

In this project, we investigate these challenges by implementing and comparing U-Net-based models with increasing levels of spatial context. Using the BraTS 2020 dataset [Menze et al., 2020], we evaluate how architectural choices affect segmentation performance across brain tumors.

2 Related Work

2.1 Convolutional Neural Networks for Segmentation

Medical image segmentation is a fundamental task in computer-aided diagnosis (CADx), enabling automated analysis of anatomical structures and pathological regions in medical imaging. Specifically, convolutional neural networks (CNNs) have demonstrated the transformative impact on segmentation accuracy and clinical utility, particularly in neuroimaging applications. [Ronneberger et al., 2015] first introduced the U-Net, an encoder-decoder architecture with skip connections that became the foundation for medical image segmentation. The architecture addresses two key challenges: limited training data and the need to preserve spatial information. This design proved particularly well-suited for medical imaging, where the U-Net achieved strong performance even with small training sets.

However, the original U-Net was designed for 2D slice-by-slice processing, which ignores volumetric context in 3D medical scans. [Özgün Çiçek et al., 2016] extended this work with 3D U-Net, which processes entire volumes using 3D convolutions and pooling operations. This captures inter-slice dependencies and anatomical continuity across all three spatial dimensions, leading to more coherent segmentation. The trade-off is substantially higher computational and memory requirements, often necessitating smaller input patches or downsampled volumes.

2.2 The BraTS Dataset

The Brain Tumor Segmentation (BraTS) challenge has driven progress in automated brain tumor analysis since 2012. [Menze et al., 2014] established the multimodal brain tumor segmentation benchmark, providing standardized protocols for data collection and evaluation. Expert radiologists annotated three tumor regions: the whole tumor, the tumor core (enhancing and necrotic regions), and the active enhancing tumor. This hierarchical labeling reflects clinical needs for treatment planning.

[Bakas et al., 2017] expanded this work by contributing TCGA glioma cases with expert segmentations and survival data. Their dataset underwent rigorous quality control, with manual corrections by board-certified neuroradiologists. It has become the standard benchmark for evaluating segmentation algorithms. We utilize the BraTS 2020 training dataset (369 cases, 240×240×155 volumes) and adhere to the same evaluation protocols. We evaluate our models using the Dice coefficient and IoU for each tumor sub-region, ensuring our results are comparable to those in previous work.

2.3 Deep Learning Approaches

[Bakas et al., 2018] analyzed submissions from the 2017-2018 BraTS challenges to identify which models worked best. Top-performing methods shared several characteristics: they utilized 3D CNNs or 2D/3D ensembles to capture volumetric information, employed U-Net variants with residual connections, applied extensive data augmentation, and employed specialized loss functions, such as Dice loss, to handle class imbalance. Notably, 3D architectures consistently outperformed 2D approaches, achieving Dice scores of 0.75-0.78 for tumor enhancement, 0.88-0.91 for the whole tumor, and 0.84-0.87 for the tumor core. The trade-off is that 3D models require much more GPU memory and training time. One example of such an approach is [Feng et al., 2018], where the authors developed a winning strategy for BraTS 2018 using an ensemble of six 3D U-Nets with different configurations. They varied the number of encoding/decoding blocks, input patch sizes, and loss function weights, then combined predictions by averaging probabilities. This ensemble achieved Dice scores of 0.754, 0.878, and 0.799 for enhancing tumor, whole tumor, and tumor core, respectively, ranking 9th overall. Their work demonstrated that ensemble methods produce more stable predictions and smoother boundaries than single models, though they noted the extreme computational cost and difficulty in determining optimal hyperparameters. They also acknowledged the "black-box" nature of deep networks, which makes it challenging to analyze what specific architectural choices contribute to performance.

Our project builds directly on these findings but takes a fundamentally different approach from competition-focused methods. Rather than maximizing performance and proposing new architectures, we systematically compare fundamental design choices: 2D, 2.5D, and 3D processing. This lets us quantify performance gains from volumetric context while documenting computational costs.

Table 1: Dataset-wide statistics for different slice ranges (368 patients). Slice counts and overall pixel class distributions.

Slice range	Slices	With Tumor	Background	Necrotic	Edema	Enhancing
(0, 155)	57,040	24,354	3,248,909,300 (98.89%)	8,129,761 (0.25%)	21,256,835 (0.65%)	7,208,104 (0.22%)
(22, 122)	37,168	22,953	2,104,937,051 (98.32%)	8,001,810 (0.37%)	20,805,289 (0.97%)	7,132,650 (0.33%)
(50, 99)	18,400	14,995	1,032,887,239 (97.46%)	6,008,080 (0.57%)	15,430,487 (1.46%)	5,514,194 (0.52%)

3 Methods

3.1 Dataset

Brain tumor segmentation is a challenging task because tumors vary significantly in their size, shape, and location. This high level of variability makes it difficult for both clinicians and automated methods to consistently identify tumor regions. The BraTS 2020 dataset helps address this challenge by providing 3D MRI scans from 369 patients, each containing four different MRI modalities: T1, T1Gd (T1 with gadolinium contrast), T2, and FLAIR. Each modality highlights different tissue characteristics: T1 shows general brain structure, T1Gd reveals areas where the contrast agent leaks into active tumor tissue, and T2 and FLAIR are particularly effective at showing fluid and swelling.

Every voxel (3D pixel) in the dataset is labeled with one of four classes: background, necrotic or non-enhancing tumor (NCR/NET), edema (ED), or enhancing tumor (ET). These labels were created manually by expert radiologists. The enhancing tumor (ET) region corresponds to actively growing tumor tissue that absorbs the contrast agent, appearing bright on T1Gd scans. In contrast, the necrotic or non-enhancing tumor (NCR/NET) region represents dead or inactive tumor tissue that does not take up contrast. Surrounding the tumor core, the edema (ED) region reflects swelling from leaking fluid from damaged blood vessels and is most clearly visible on FLAIR and T2 images.

Because each of these tissue types has distinct visual characteristics and appears differently across MRI modalities, using all four sequences together provides the most complete picture of the tumor. This multimodal information is essential for training deep learning models to segment tumors accurately.

3.2 Pre-processing

All MRI volumes were used at their original resolution. No global intensity clipping or histogram-based standardisation was applied. Instead, each 2D slice underwent per-slice min-max normalisation:

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}.$$

To exclude empty superior and inferior regions, both the 2D and 2.5D models were trained only on axial slices 22–122. This range nearly retains all tumour-containing anatomy and the overall class distribution while reducing computational cost (see Table 1). A smaller range could result in a much different class distribution. This could influence the performance of data that is never seen during training. We used all slices within this range.

For the 2.5D model, a limited volumetric context was introduced by aggregating neighbouring slices. For each central slice, two slices above and two below were averaged for each modality, producing an input tensor with four channels (one per modality). The 2D model used only the central slice for each modality. For the 3D U-Net, we used the entire MRI volume (all 155 axial slices) without restricting the slice range. Training was performed using full MRI volumes to maximize global context. Due to the spatial dimensions (240×240×155), inputs were padded to ensure dimensions were divisible by the network’s downsampling factor, rather than extracting smaller sub-patches.

Finally, we split the dataset at the patient level into 70% for training, 15% for validation, and 15% for testing. We ensured that no slices from the same patient appeared in different splits, eliminating patient-specific information leakage and supporting a fair evaluation of model generalisation.

3.3 Model Architectures

We evaluate three segmentation models to study the influence of spatial context.

2D U-Net

The 2D U-Net follows the classical encoder–decoder structure with skip connections. Each level contains two 3×3 convolutions, Batch Normalisation, and ReLU activation. The number of channels increases as:

$$[64, 128, 256, 512, 1024]$$

Downsampling is performed via max pooling and upsampling via transposed convolutions. No dropout layers were used. The network takes a four-channel input corresponding to the four MRI modalities.

2.5D U-Net

The 2.5D U-Net is architecturally identical to the 2D model but operates on 2D slices that encode additional volumetric context via the slice-aggregation procedure described above. This strategy enhances performance by incorporating local 3D structure while retaining the computational efficiency of 2D convolutions.

3D U-Net

The 3D model replaces all 2D operations with their 3D counterparts (Conv3D, MaxPool3D). To maximize spatial context, we utilized a Full Volumetric training strategy rather than patch-based extraction. However, processing full $240 \times 240 \times 155$ volumes incurs a massive memory footprint. To accommodate this within the available VRAM, we were forced to constrain the model’s width: whereas the 2D and 2.5D architectures start with 64 feature channels, the 3D U-Net was configured with a reduced capacity, starting at 32 feature channels and scaling to a maximum of 512. This results in a significantly smaller parameter space compared to the slice-based variants.

Normalization and Activation

Across all architectures (2D, 2.5D, and 3D), each convolutional layer was immediately followed by **Batch Normalization** to stabilize learning and a ReLU activation function. This ensures that architectural comparisons remained focused on spatial dimensionality rather than variations in layer composition.

3.4 Loss Function

Brain tumour segmentation in BraTS is characterised by extreme class imbalance, with background voxels accounting for nearly 99% of all voxels and enhancing tumour regions comprising less than 0.3%. To address this imbalance, we trained all models using a combined Dice and Cross-Entropy loss:

$$\mathcal{L} = 0.5 \cdot \mathcal{L}_{\text{CE}} + 0.5 \cdot \mathcal{L}_{\text{Dice}}.$$

Cross-Entropy contributes stable voxel-wise gradients during early training, while Dice loss optimises region-level overlap and prevents the model from collapsing into majority-class predictions. This hybrid loss function has become standard in medical image segmentation and was essential for stable optimisation in our experiments.

3.5 Training Setup

All experiments were executed on the *Accelgor* cluster of the Ghent University HPC infrastructure^[1]. Accelgor consists of 9 compute nodes, each equipped with dual 24-core AMD EPYC 7413 (Milan, 2.2 GHz) processors and four NVIDIA Ampere A100 GPUs, each providing 80 GB of GPU memory. Although each node provides substantial CPU and GPU resources, all training in this project was performed on a single Accelgor node, using only one GPU, 6 CPU cores, and 52.7 GiB of RAM.

3.6 Evaluation Metrics

Evaluation followed the official BraTS protocol, using the Dice coefficient and Intersection-over-Union (IoU) to quantify segmentation quality. Since our model outputs four voxel-wise classes (background, necrotic/non-enhancing tumor, edema, and enhancing tumor), we report per-class Dice and IoU scores for NCR/NET, Edema, and Enhancing Tumor. These classes correspond directly to the clinically relevant BraTS regions and allow us to assess the model’s ability to distinguish between the different tumor subcomponents. We computed overall accuracy and mean Dice to provide a global measure of segmentation performance. In addition to quantitative evaluation, we performed qualitative visual inspection to analyse segmentation behaviour, boundary sharpness, and failure modes across architectures.

<https://docs.hpc.ugent.be/Linux/infrastructure/#interactive-debug-cluster>

4 Results

This section presents the quantitative and qualitative performance of the 2D, 2.5D, and 3D U-Net architectures. We analyze the training dynamics, compare the varying spatial contexts, and visualize specific success and failure cases.

4.1 Training Dynamics

We monitored the learning progression using the combined Dice and Cross-Entropy loss. Figure 1 illustrates the training and validation curves for all three architectures.

The 2D U-Net (Batch Size 64) showed stable convergence. The training loss decreased rapidly from an initial 0.60 to 0.13 by epoch 9. The validation Dice score improved consistently, starting at 0.64 and peaking at approximately 0.74.

The 2.5D U-Net (Batch Size 16) exhibited the fastest convergence and highest stability. As seen in Figure 1 (center), the Dice score jumped significantly within the first two epochs (from 0.66 to 0.77) and plateaued near 0.80 by epoch 5. This suggests that including local volumetric context via slice aggregation provides a strong, immediate signal for tumor delineation.

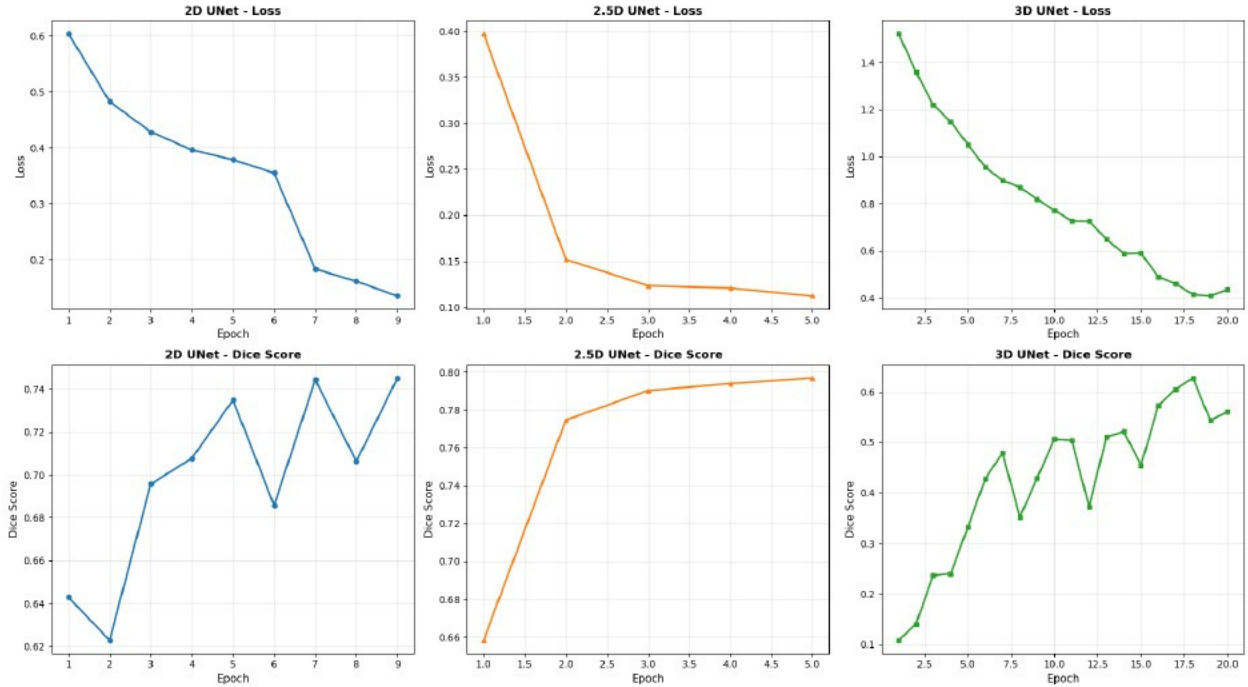


Figure 1: Training history for 2D, 2.5D, and 3D architectures. The 2.5D model (center) achieves the highest Dice score (≈ 0.80) with the lowest loss. The 3D model (right) shows significantly higher loss magnitude and slower metric improvement.

4.2 Comparison with 3D U-Net

The 3D U-Net demonstrated significantly different learning dynamics compared to the slice-based approaches. While the model did learn—improving its validation Dice score from 0.10 to approximately 0.56 over 20 epochs—it failed to match the performance of the 2D variants.

The 3D model’s training was characterized by:

1. **Higher Loss Magnitude:** The loss remained above 0.40 even after 20 epochs, compared to ≈ 0.11 for the 2.5D model.
2. **Slower Convergence:** It required more epochs to reach a Dice score of 0.50 than the 2D models required to reach 0.70.

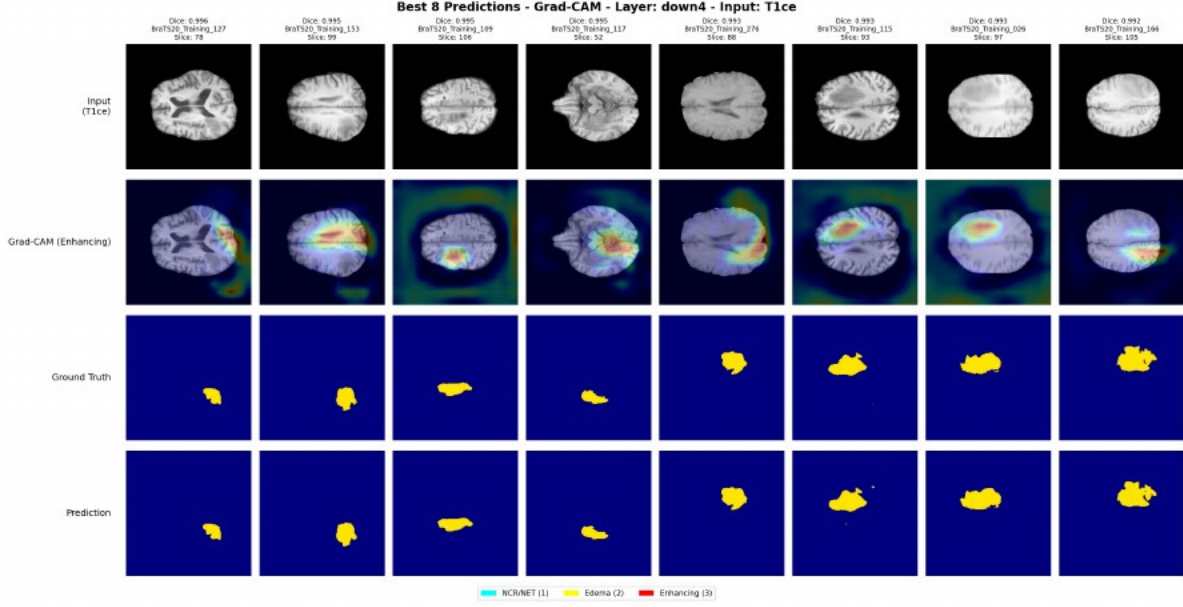


Figure 2: Grad-CAM visualization of the best eight predictions (with dice-score < 1 to prevent getting only samples without tumor slices) for enhancing tumor segmentation (class 3) using the bottleneck layer (down4) with T1ce MRI input on the test set. The first row shows the original T1ce input images from different patients. The second row displays the Grad-CAM heatmaps overlaid on the input images, highlighting the regions most relevant to the model’s decision-making process for identifying and enhancing tumor tissue. Red regions indicate high activation (strong model focus), while blue regions indicate low activation. The third row presents the ground truth segmentation masks, and the fourth row shows the model’s predictions.

3. **Instability:** The validation curves (Figure 1 right) exhibit higher volatility between epochs compared to the smooth progression of the 2.5D model.

4.3 Quantitative Comparison

Table 2 summarizes the performance on the test set. The 2.5D U-Net achieved the highest performance across all metrics, validating the efficacy of the slice-aggregation method. The 3D U-Net, limited by GPU memory constraints, which necessitated a reduced channel capacity (starting at 32 filters vs 64 for 2D) and smaller batch sizes, underperformed the baseline.

Table 2: Comparison of model performance on the test set. Note: 3D U-Net channel capacity was reduced to fit GPU memory constraints, resulting in significantly fewer parameters.

Model	Params	Epochs	Batch	Loss	Dice _{avg}	Dice _{bg}	Dice _{nec}	Dice _{ede}	Dice _{enh}
2D U-Net	31.0M	9	64	0.134	0.745	0.998	0.599	0.661	0.754
2.5D U-Net	31.0M	5	16	0.113	0.797	0.998	0.728	0.722	0.792
3D U-Net	22.6M	20	2	0.408	0.560	0.993	0.505	0.547	0.543

4.4 Qualitative Results

To interpret the model’s decision-making, we utilized Grad-CAM visualization. Figure 2 illustrates a successful segmentation case by the 2.5D model. The heatmaps confirm that the model focuses specifically on the hyper-intense regions in the T1ce modality when segmenting enhancing tumors, matching clinical expectations.

Conversely, Figure 3 highlights failure modes. In these cases, the model often struggles with discontinuities in tumor boundaries or fails to detect small, isolated tumor islands. Table 3 details the pixel-level confusion, revealing that the necrotic core (Class 1) remains the most challenging region to segment, yielding the highest false negative rate relative to its size.

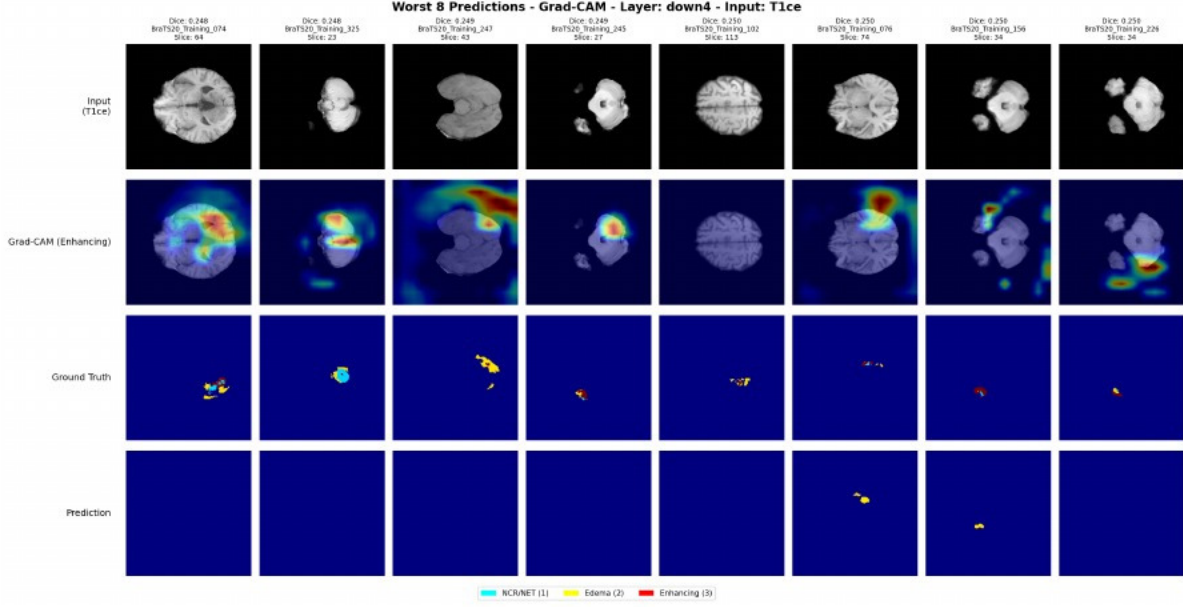


Figure 3: Grad-CAM visualization of worst-performing predictions on the test set, revealing model failure modes. The model frequently mislocates attention to non-tumor regions, fails to detect small enhancing tumors, or produces false positive predictions. These cases demonstrate challenges with edge slices, small tumor volumes, and boundary artifacts (layer down4, T1ce input, enhancing tumor class).

Table 3: Class-wise precision, recall, and pixel-level confusion counts for the 2.5D U-Net on the test set.

Class	Precision	Recall	TP	FP	FN
Background (0)	0.9985	0.9984	316,706,394	462,283	502,353
Necrotic (1)	0.7643	0.6735	802,462	247,444	389,057
Edema (2)	0.7987	0.8001	2,565,259	646,513	640,975
Enhancing (3)	0.7377	0.8739	833,305	296,340	120,195

5 Discussion

5.1 Impact of Spatial Context

The results highlight a nuanced relationship between spatial context and segmentation accuracy. The 2.5D U-Net outperformed both the 2D baseline and the 3D model, suggesting that aggregating local adjacent slices captures sufficient volumetric information for boundary definition without the optimization difficulties encountered in our full 3D implementation.

5.2 Optimization Challenges in 3D Architectures

Despite the theoretical benefits of volumetric processing, the 3D U-Net achieved the lowest Dice scores. This result was not caused by insufficient computational resources, but rather by optimization challenges related to our data representation and normalization strategies:

1. **Full Volumetric Strategy vs. Batch Size:** We opted for a **Full Volumetric** training strategy to maximize global context, rather than a patch-based approach. Processing entire $240 \times 240 \times 155$ volumes incurs an immense memory footprint during backpropagation. Even with high-performance GPU resources, this strategy restricted the batch size to $N = 2$.
2. **Batch Normalization Failure:** The enforced small batch size proved detrimental to the **Batch Normalization** layers used throughout the network. Batch Norm relies on accurate statistical estimation of the mean and variance across the batch. With only two samples per batch, these estimates became highly noisy, leading to optimization instability and undertraining, as shown in Figure 1.

3. **Model Capacity Trade-off:** To accommodate the full volumes within memory, we were forced to reduce the 3D model’s filter capacity (starting at 32 filters). This resulted in a model with approximately **22.6 million parameters**, significantly fewer than the **31.0 million parameters** of the 2D and 2.5D architectures. Consequently, the 3D model was tasked with learning a more complex volumetric representation with $\approx 27\%$ less representational capacity, which directly contributed to its inability to capture complex tumor morphology.

5.3 Undertraining

The training curves (Figure 1) confirm that the 3D model suffered from optimization difficulties. While the 2D and 2.5D models converged rapidly, the 3D model’s loss remained high, and the Dice score improved linearly without reaching a plateau. This indicates that the chosen configuration (Full Volume + Batch Norm) created a learning landscape that was difficult to traverse efficiently, requiring significantly more epochs or a different normalization strategy to converge.

5.4 Class-Specific Challenges

Consistent with previous BraTS challenges, distinguishing between tumor sub-regions proved more difficult than the binary tumor/background segmentation. The 2.5D model achieved strong overall Dice scores but struggled specifically with necrotic tissue (NCR/NET), which had the lowest recall and the highest false-negative rate (Table 3).

Qualitative analysis via Grad-CAM (Figures 2 and 3) suggests that the model sometimes confuses necrotic cores with the surrounding edema or non-enhancing tumor, likely due to their similar intensity profiles in T1 and T2 modalities. Future work could address this by employing class-aware loss functions (e.g., Focal Tversky Loss) to penalize errors in these smaller, harder-to-classify subregions more heavily.

5.5 Limitations

This study faced specific constraints regarding architectural optimization and training strategies. The most significant limitation was the decision to utilize a **Full Volumetric** training strategy for the 3D U-Net. While theoretically optimal for context, the substantial memory cost necessitated a reduced model capacity and a batch size of $N = 2$. This small batch size proved incompatible with standard Batch Normalization, leading to noisy statistical estimates and optimization instability during training. Future implementations could resolve this by adopting patch-based sampling to increase batch sizes or by replacing Batch Normalization with Group or Instance Normalization, which are independent of batch statistics.

Additionally, our general training pipeline employed minimal data augmentation and a standard hybrid loss function. The mild overfitting observed in the 2D and 2.5D models suggests that aggressive geometric and elastic deformations are necessary to improve generalization. Furthermore, while the Dice and Cross-Entropy losses addressed general class imbalance, they struggled with the smallest subregions; class-aware objectives such as Focal or Tversky losses could reduce the false-negative rates observed in necrotic tissue segmentation.

5.6 Conclusion

This project investigated the architectural trade-off between spatial context and computational constraints in brain tumor segmentation. Our results identify the 2.5D U-Net as the effective "sweet spot" for medical image analysis. By aggregating local volumetric context, it achieves superior boundary delineation compared to 2D baselines while retaining training stability. Conversely, our findings show that adopting 3D architectures alone does not guarantee performance. Without specific optimization strategies—such as patch-based processing or specialized normalization layers, 3D models are prone to convergence failure. Consequently, 2.5D architectures offer a pragmatic and robust alternative for clinical applications where computational resources and model reliability are paramount.

6 Resources

- **Dataset:** BraTS 2020 dataset. Available at: <https://www.kaggle.com/datasets/awsaf49/brats20-dataset-training-validation>

References

Andrea Comba, Syed Faisal, María Luisa Varela, Todd Hollon, Wajid Al-Holou, Yoshie Umemura, Felipe Nunez, Sebastien Motsch, Maria Castro, and Pedro Lowenstein. Uncovering spatiotemporal heterogeneity of high-

- grade gliomas: From disease biology to therapeutic implications. *Frontiers in Oncology*, 11, 08 2021. doi:[10.3389/fonc.2021.703764](https://doi.org/10.3389/fonc.2021.703764)
- Kaiming He, X. Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 770–778, 2015. URL <https://api.semanticscholar.org/CorpusID:206594692>
- Gao Huang, Zhuang Liu, and Kilian Q. Weinberger. Densely connected convolutional networks. *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2261–2269, 2016. URL <https://api.semanticscholar.org/CorpusID:9433631>
- Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation, 2015. URL <https://arxiv.org/abs/1505.04597>
- Aston Zhang, Zachary Lipton, Mu Li, and Alexander Smola. Dive into deep learning, 06 2021.
- X. X. Yin, L. Sun, Y. Fu, R. Lu, and Y. Zhang. U-net-based medical image segmentation. *Journal of Healthcare Engineering*, 2022:4189781, April 2022. doi:[10.1155/2022/4189781](https://doi.org/10.1155/2022/4189781) Retracted in: J Healthc Eng. 2023 Oct 18;2023:9890389. doi:[10.1155/2023/9890389](https://doi.org/10.1155/2023/9890389).
- Bjoern Menze, Andras Jakab, Stefan Bauer, Jayashree Kalpathy-Cramer, Keyvan Farahani, Justin Kirby, Yuliya Burren, Nicole Porz, Johannes Slotboom, Roland Wiest, et al. Brats 2020: Multimodal brain tumor segmentation challenge dataset. <https://www.med.upenn.edu/cbica/brats2020/data.html>, 2020. Data accessed via Kaggle mirror: <https://www.kaggle.com/datasets/awsaf49/brats20-dataset-training-validation>
- Özgün Çiçek, Ahmed Abdulkadir, Soeren S. Lienkamp, Thomas Brox, and Olaf Ronneberger. 3d u-net: Learning dense volumetric segmentation from sparse annotation, 2016. URL <https://arxiv.org/abs/1606.06650>
- Bjoern Menze, András Jakab, Stefan Bauer, Jayashree Kalpathy-Cramer, Keyvan Farahani, Justin Kirby, Yuliya Burren, Nicole Porz, Johannes Slotboom, Roland Wiest, Levente Lencsényi, Elizabeth Gerstner, Marc-André Weber, Tal Arbel, Brian Avants, Nicholas Ayache, Patricia Buendia, Louis Collins, Nicolas Cordier, and Koen Van Leemput. The multimodal brain tumor image segmentation benchmark (brats). *IEEE Transactions on Medical Imaging*, 99, 12 2014. doi:[10.1109/TMI.2014.2377694](https://doi.org/10.1109/TMI.2014.2377694)
- Spyridon Bakas, Hamed Akbari, Aristeidis Sotiras, Michel Bilello, Martin Rozycki, Justin Kirby, John Freymann, Keyvan Farahani, and Christos Davatzikos. Advancing the cancer genome atlas glioma mri collections with expert segmentation labels and radiomic features. *Scientific Data*, 4, 09 2017. doi:[10.1038/sdata.2017.117](https://doi.org/10.1038/sdata.2017.117)
- Spyridon Bakas, Mauricio Reyes, András Jakab, Stefan Bauer, Markus Rempfler, Alessandro Crimi, Russell Takeshi Shinohara, Christoph Berger, Sung Min Ha, Martin Rozycki, Marcel Prastawa, Esther Alberts, Jana Lipková, John B. Freymann, Justin S. Kirby, Michel Bilello, Hassan M. Fathallah-Shaykh, Roland Wiest, Jan Stefan Kirschke, Benedikt Wiestler, Rivka R. Colen, Aikaterini Kotrotsou, Pamela J. LaMontagne, Daniel S. Marcus, Mikhail Milchenko, Arash Nazeri, Marc-André Weber, Abhishek Mahajan, Ujjwal Baid, Dongjin Kwon, Manu Agarwal, Mahbubul Alam, Alberto Albiol, Antonio Albiol, Alex Varghese, Tran Anh Tuan, Tal Arbel, Aaron Avery, B. Pranjali, Subhashis Banerjee, Thomas Batchelder, Nematollah K. Batmanghelich, Enzo Battistella, Martin Bendszus, Eze Benson, José Bernal, George Biros, Mariano Cabezas, Siddhartha Chandra, Yi-Ju Chang, and et al. Identifying the best machine learning algorithms for brain tumor segmentation, progression assessment, and overall survival prediction in the brats challenge. *ArXiv*, abs/1811.02629, 2018. URL <https://api.semanticscholar.org/CorpusID:53225711>
- Xue Feng, Nicholas Tustison, and Craig Meyer. Brain tumor segmentation using an ensemble of 3d u-nets and overall survival prediction using radiomic features, 2018. URL <https://arxiv.org/abs/1812.01049>